

Cognitive Radio Networks

Advanced Mobile Communication Networks



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Introduction

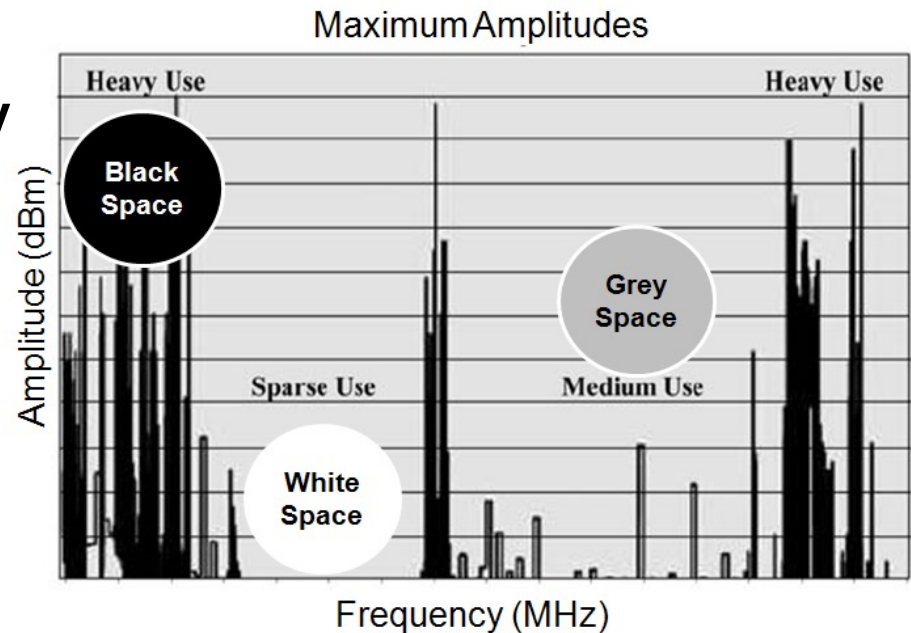
Questions to be answered:

- What is the idea of cognitive radio? Why is it important?
- What defines a primary network, what a secondary network?
- What defines a spectrum hole?
- What are the main tasks and challenges of cognitive radios?
- Why do cognitive radios need to be configurable?
- What are the differences between cognitive radio networks and traditional wireless networks?

Introduction

- Limited spectrum
- Inefficiency in spectrum usage
 - Utilization of the assigned licensed spectrum varies between 15% and 85%
- A new technology is required to enable better utilization of unused licensed spectrum

→ **Cognitive radio technology**



What is a Cognitive Radio?

- Definitions
 - **Federal Communications Commission Definition:** a cognitive radio is a radio that can change its transmitter parameters based on interaction with the environment in which it operates
 - **Mitola Definition:** the cognitive radio identifies the point at which wireless PDAs and the related networks are sufficiently computationally intelligent on the subject of radio resources and related computer-to-computer communications
 - to detect user communication needs as a function of use context, and
 - to provide radio resources and wireless services most appropriate to those needs

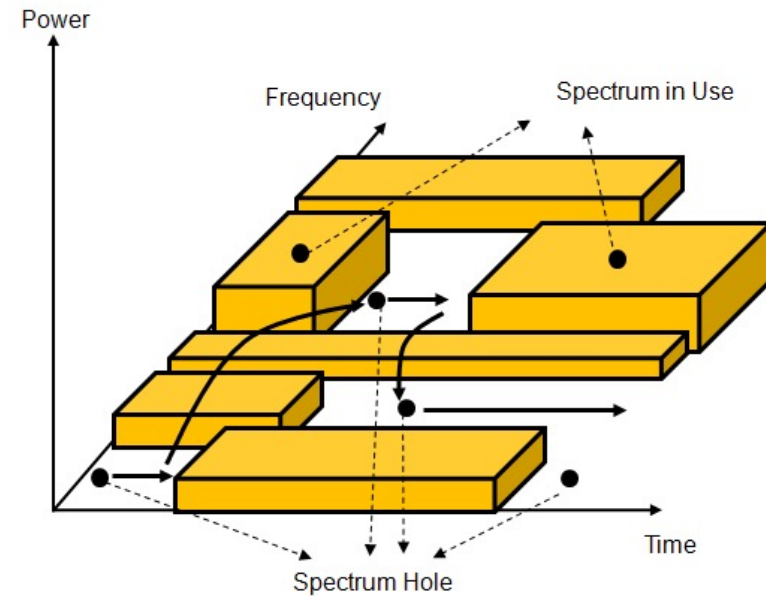
What is a Cognitive Radio?

- Characteristics
 - **Cognitive capability**: the capability of the radio technology of capturing or sensing information from its radio environment (monitoring the power in some frequency band)
 - **Reconfigurability**: the ability of the radio to be dynamically programmable according to the radio environment (transmission and receipt of data on a variety of frequencies and using various radio access technologies)

= Learn + adapt

What Should Cognitive Radio Enable to Do?

- Determination of unused portion of spectrum (spectrum holes or white space)
- Selection of the best available channel
- Coordination of the access with other users
- Detection of the appearance of licensed users → vacate the channel



Cognitive Radio Terminology

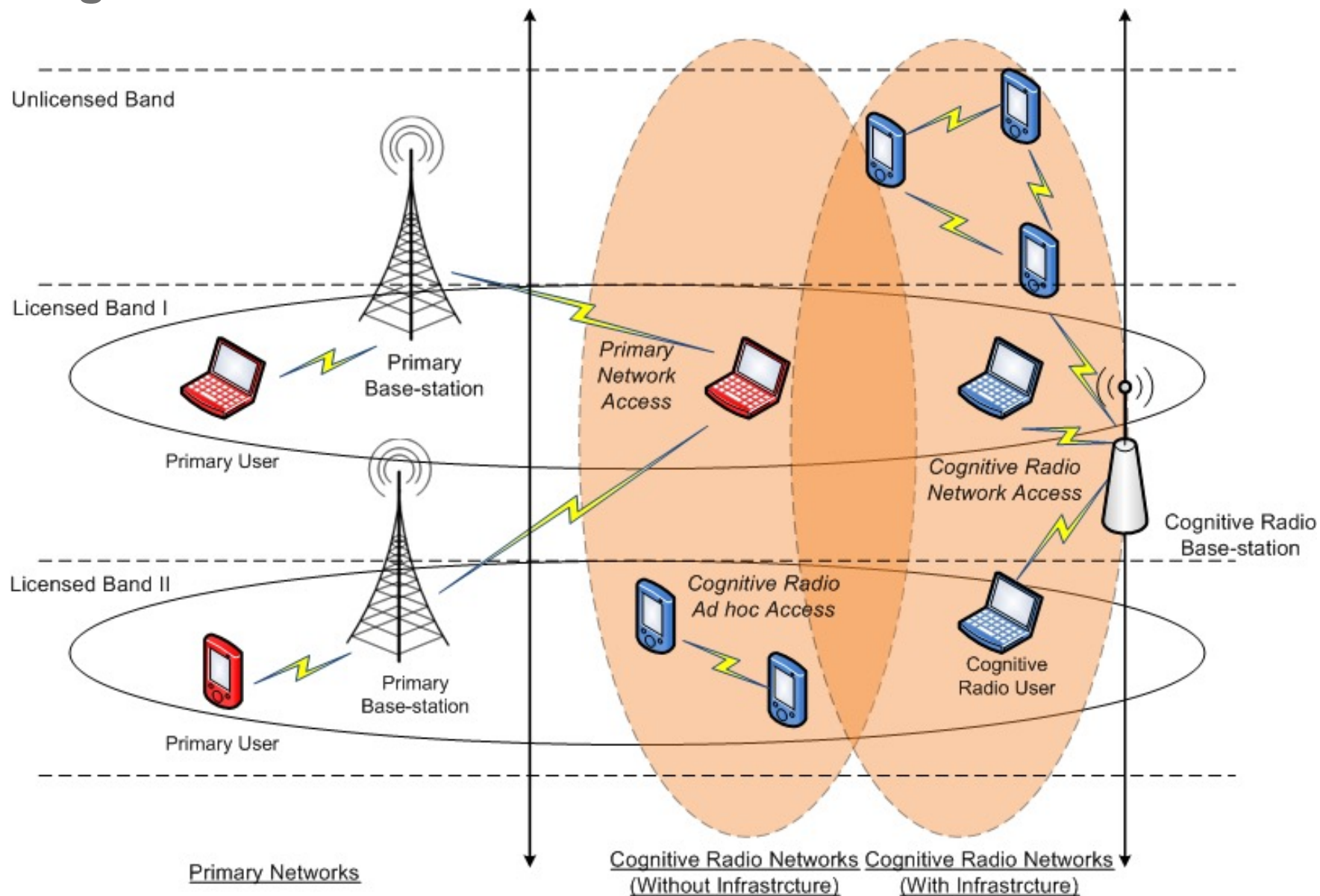
- **Primary networks**

- Networks with access right to certain spectrum bands, e.g. common cellular systems and TV broadcast networks
- Users of these networks are referred to as **primary users**. They have the right to operate in licensed spectrum.
- Users of a certain primary network do not care of other primary or secondary network users

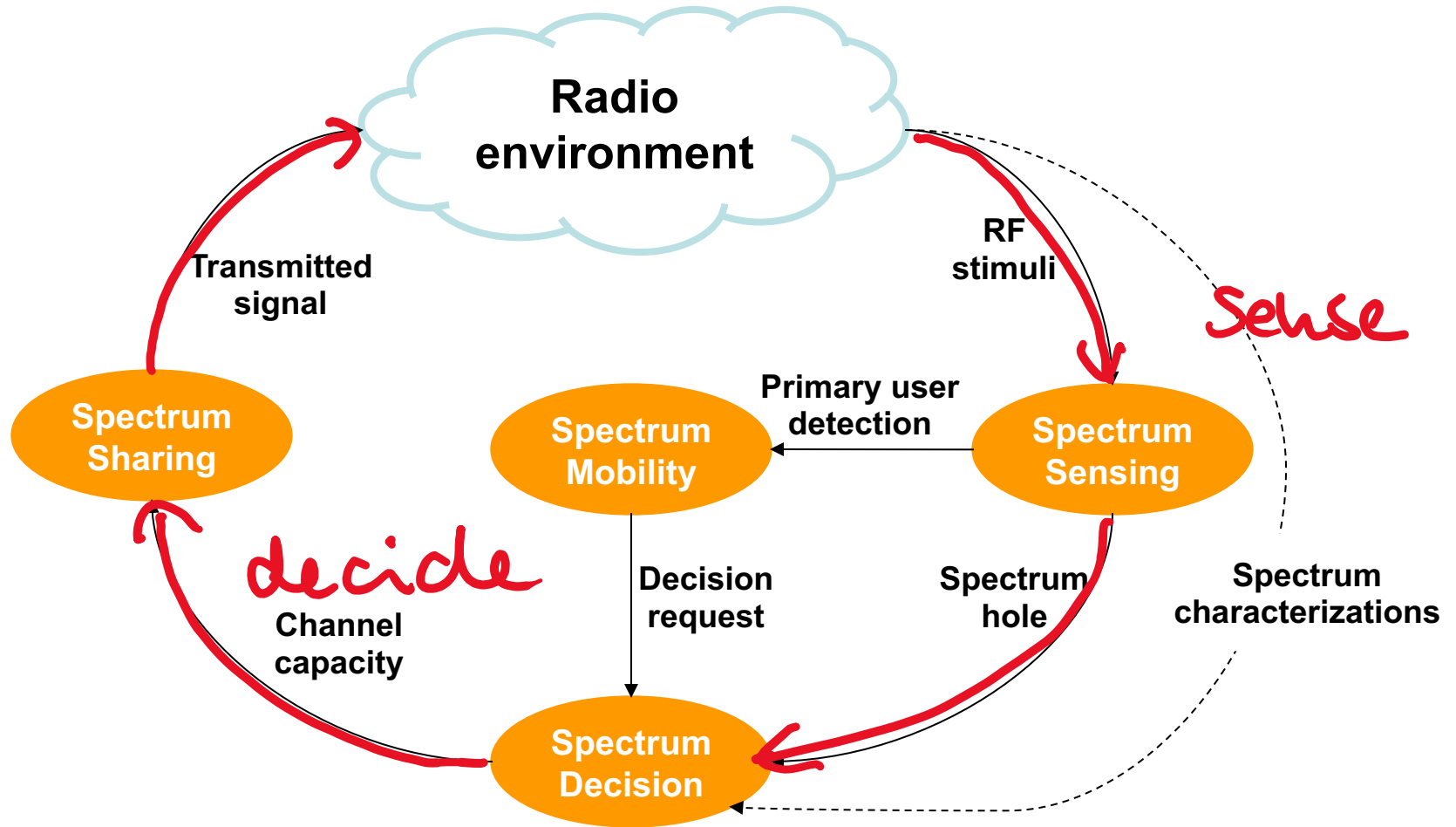
- **Secondary networks**

- Do not have license to operate in the spectrum band
- Users of these networks are referred to as **secondary users**. They have no right to access licensed bands currently used.
- This is also called Opportunistic Spectrum Access or Dynamic Spectrum Access – DSA
- Additional functionalities are required to share licensed spectrum bands with other secondary or primary networks

Cognitive Radio Network Architecture



Cognition Cycle *of secondary user*



Spectrum Sensing

Questions to be answered:

- Why is accurate spectrum sensing for cognitive radios important?
- Explain the hidden terminal problem for cognitive radio networks
- What are the advantages of cooperative sensing? How does cooperative sensing solve the hidden terminal problem?
- What is the idea of matched filter sensing? What are the prerequisites?
- What does energy detection mean? What is its advantage?
- How does cyclostationary feature detection work?
- What is the idea of interference temperature management?

Spectrum Sensing

Task of sensing:

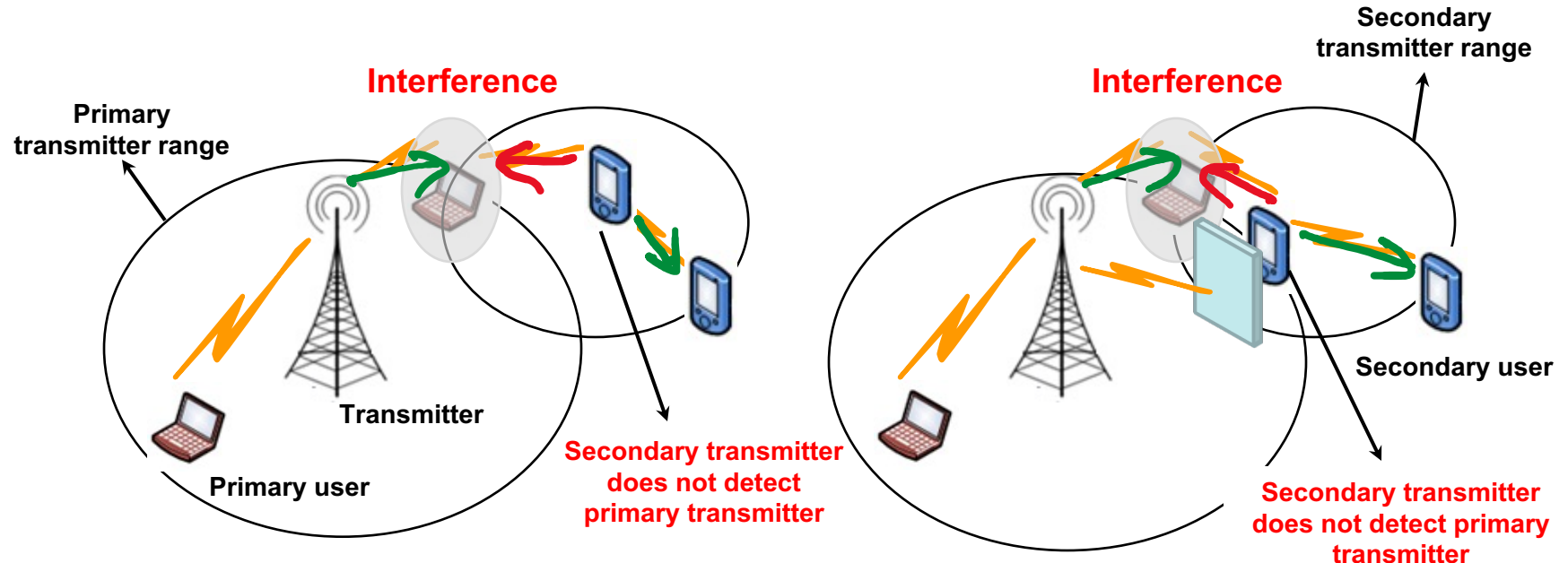
- Identification of spectrum holes for which disturbance of primary receivers (interference) is prevented, i.e. to avoid the use of spectrum where the primary network is impacted by interference

Challenge:

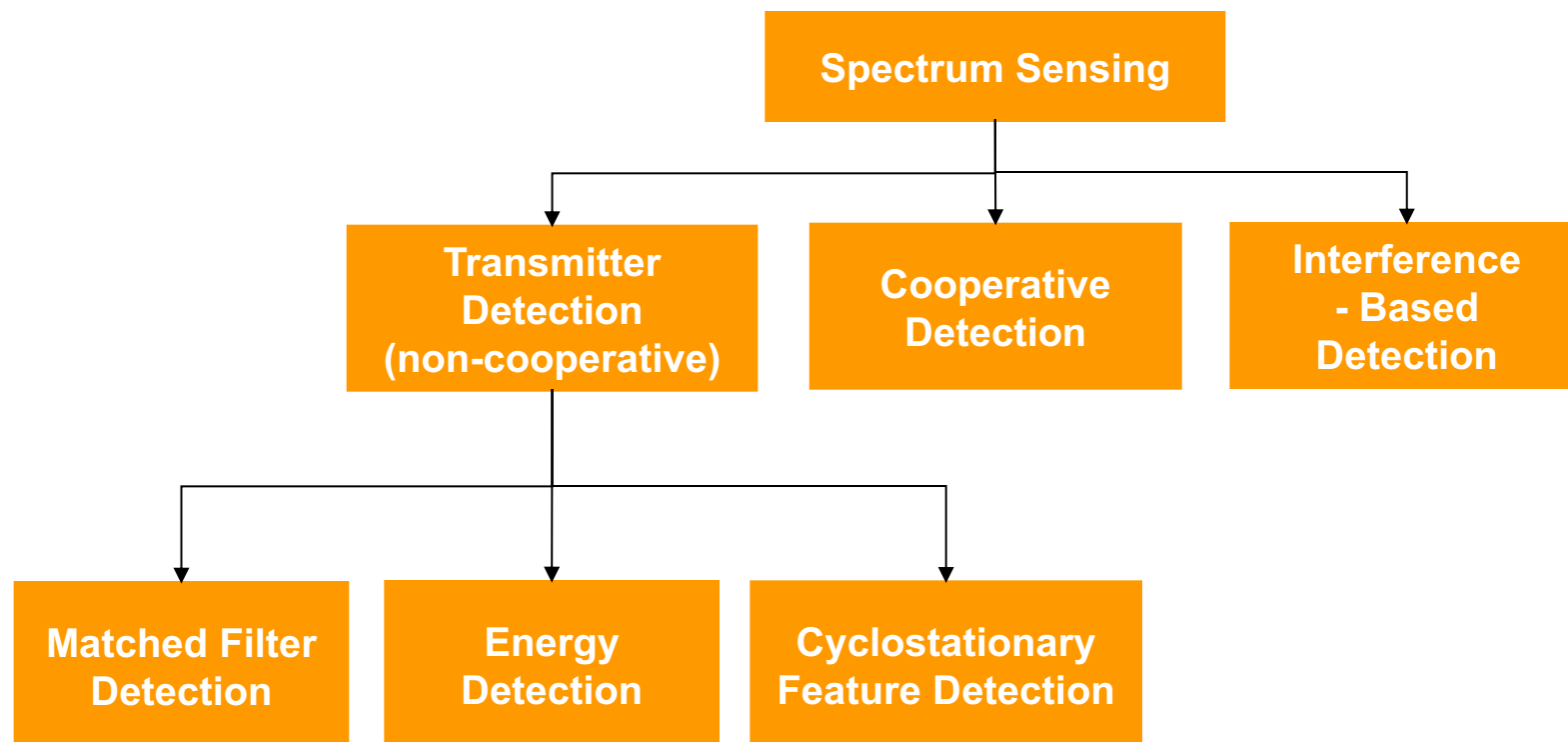
- Secondary user can only sense the primary transmitters, while interference occurs at the primary receiver (hidden-terminal problem)

The Problem of Primary Transmitter Detection

- Primary and secondary networks are physically separated but share the same radio resource → transmitter detection can not avoid interference due to the lack of the primary receiver's information
- Even if the secondary user has a line of sight to the primary receiver, the secondary user may not be able to detect the primary transmitter, e.g. due to shadowing, etc.
- Cannot prevent hidden terminal problem



Spectrum Sensing Techniques



As an alternative to sensing, the spectrum available for secondary use may also be provided by a geolocation database, e.g. for (temporarily) unused TV bands

Transmitter Detection

- Matched filter detection
 - Check of correlation of received signal with a given template, e.g. a pilot, that allows detecting the presence of a known signal (maximize SNR)
 - Optimal detector when all characteristics of the primary user signal are known, otherwise poor performance
 - High complexity due to need for carrier synchronization
- Cyclostationary feature detection
 - Mean and autocorrelation of modulated signals exhibit periodicity (modulated signals are coupled normally with sine wave carriers, pulse trains, repeating spreading, etc.)
 - Differentiates noise from modulated signal (noise is a wide-sense stationary signal with no correlation)
 - Computationally complex and requires long observation time
- Energy detection
 - Used if primary user information is unknown

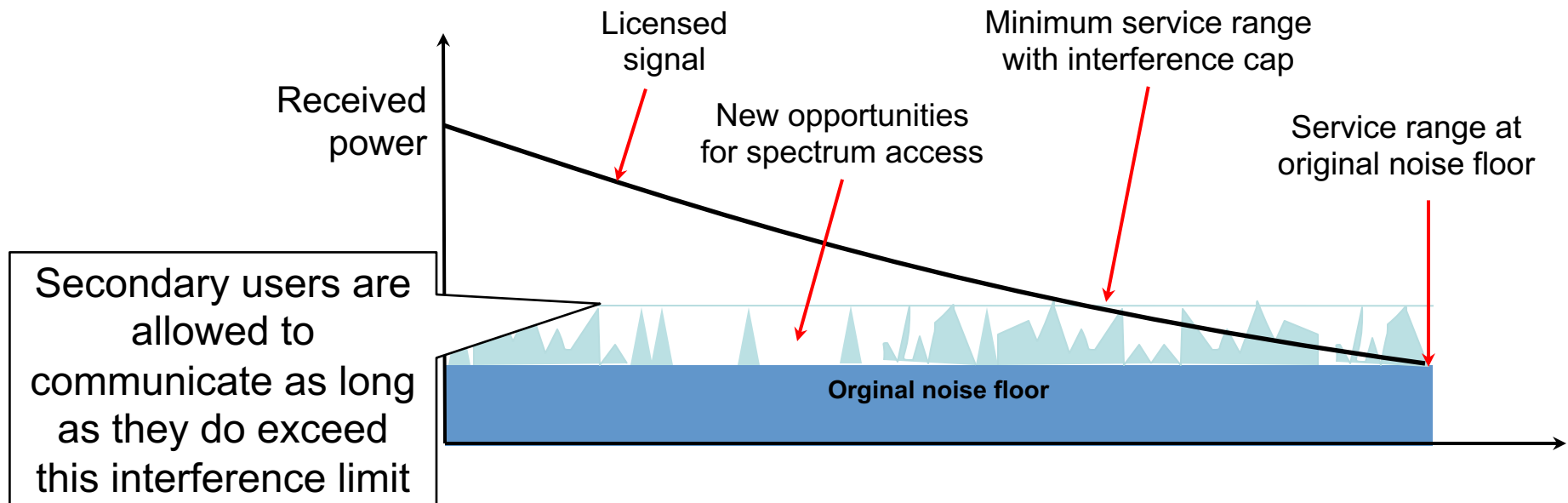
For details see: H. Sun, A. Nallanathan, C. Wang and Y. Chen, "Wideband spectrum sensing for cognitive radio networks: a survey," in *IEEE Wireless Communications*, vol. 20, no. 2, pp. 74-81, April 2013

Cooperative Detection

- Incorporating information from multiple secondary users to detect primary users
- Cooperative detection strategies
 - Centralized detection
 - Controlled mostly by cognitive BSs
 - Cognitive BS collects sensing information from all secondary users it serves and detects spectrum holes
 - Distributed detection
 - No centralized infrastructure
 - Observations are exchanged among secondary users
- More accurate detection than that based on single secondary user observations, e.g. transmitter detection

Interference-Based Detection (Interference Temperature Management)

- Interference takes place at the receiver side (i.e. the primary receiver is disrupted), but is controlled by the transmitter side, e.g. using power control, etc.
- Interference temperature management regulates received power at primary user instead of transmitted power, typically approximated by maximum transmit power for secondary users



Challenges

- Interference temperature management
 - Secondary users are aware of their locations and transmission power, but they are not aware of the interference temperature at neighboring primary users
- Spectrum sensing in multi-user networks
 - Multi-user environments (secondary networks coexisting with each other as well as with primary networks) make it more difficult to sense primary users
 - Cooperative detection can be considered as possible solution for such environments
- Detection capability
 - Detection (and analysis) of primary users in very short time is essential especially if the primary network has highly dynamic behavior

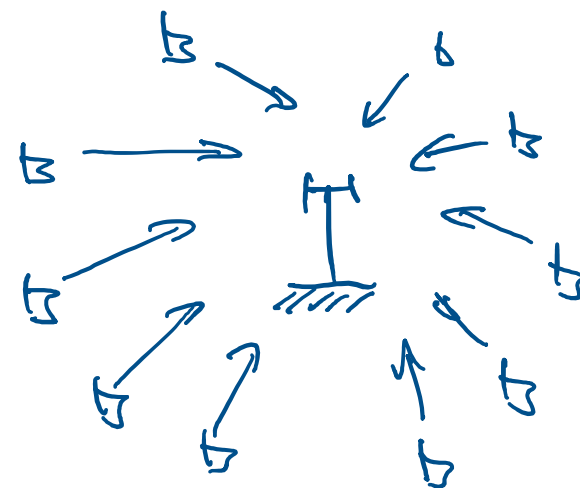
Spectrum Decision, Sharing and Mobility

Questions to be answered:

- What are the advantages of a centralized secondary network architecture?
- What are the differences between overlay and underlay?
- Which criteria are relevant for the effective selection of the spectrum hole?
- What is spectrum mobility?
- When can a secondary transmitter use a frequency band at the same time as a primary transmitter? What are the requirements for this?
- For which kind of cognitive radio approaches is the application of spectrum spreading and CDMA especially useful? What is the advantage?

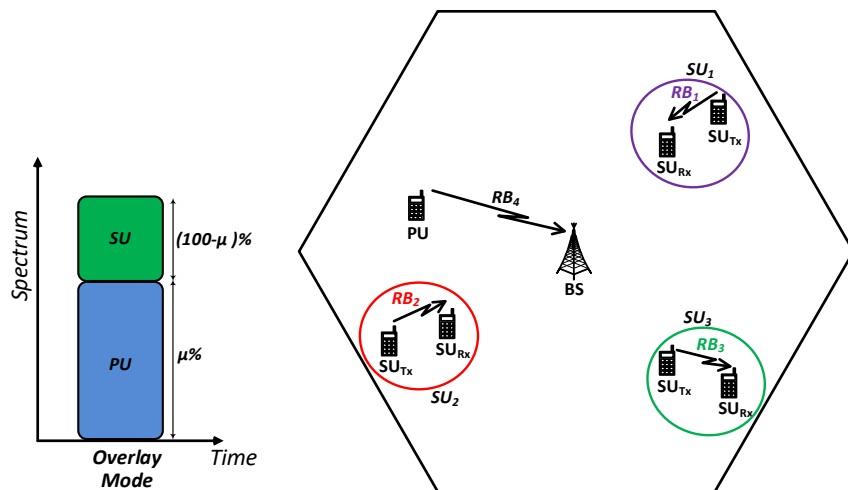
CRN Architecture

- Centralized
 - Global knowledge and centralized control of the spectrum allocation and access
 - Secondary users report observations to the centralized entity
- Distributed/self-organized
 - Decide locally based on local knowledge
 - Learn from the environment (and the neighbors)
 - Cooperative vs. non-cooperative (selfish)
 - Observation of each node are shared (or not) with other nodes
 - Cooperation provides better spectrum utilization at the cost of signaling overhead

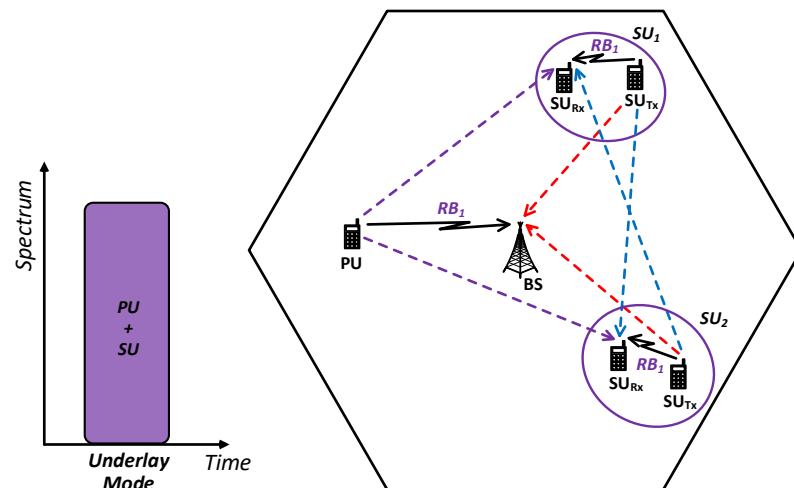


Spectrum Sharing – Overlay vs. Underlay

Overlay spectrum sharing



Underlay spectrum sharing



- Overlay spectrum sharing
 - Secondary users access unused spectrum holes
→ Avoid interference to primary users
- Underlay spectrum sharing
 - Secondary users stay within acceptable interference temperature
 - Often based on CDMA and spread spectrum techniques

Spectrum Decision

Issues to be considered for spectrum decision:

- Spectrum bands are spread over a wide frequency range possibly including licensed and unlicensed bands
- Radio environments may show fast and unpredictable variation over time (between secondary users as well as towards primary users)
- Primary user behavior may vary highly for different spectrum and different radio technologies
- Allowable interference (or transmit power) may vary for different spectrum due to radio technology, primary user locations, environmental conditions with impact on pathloss, etc.

Task:

- Secondary users have to select most suitable spectrum band meeting their specific QoS requirements (along with the requirements of the primary network)

Spectrum Decision – Characteristics of Spectrum Holes

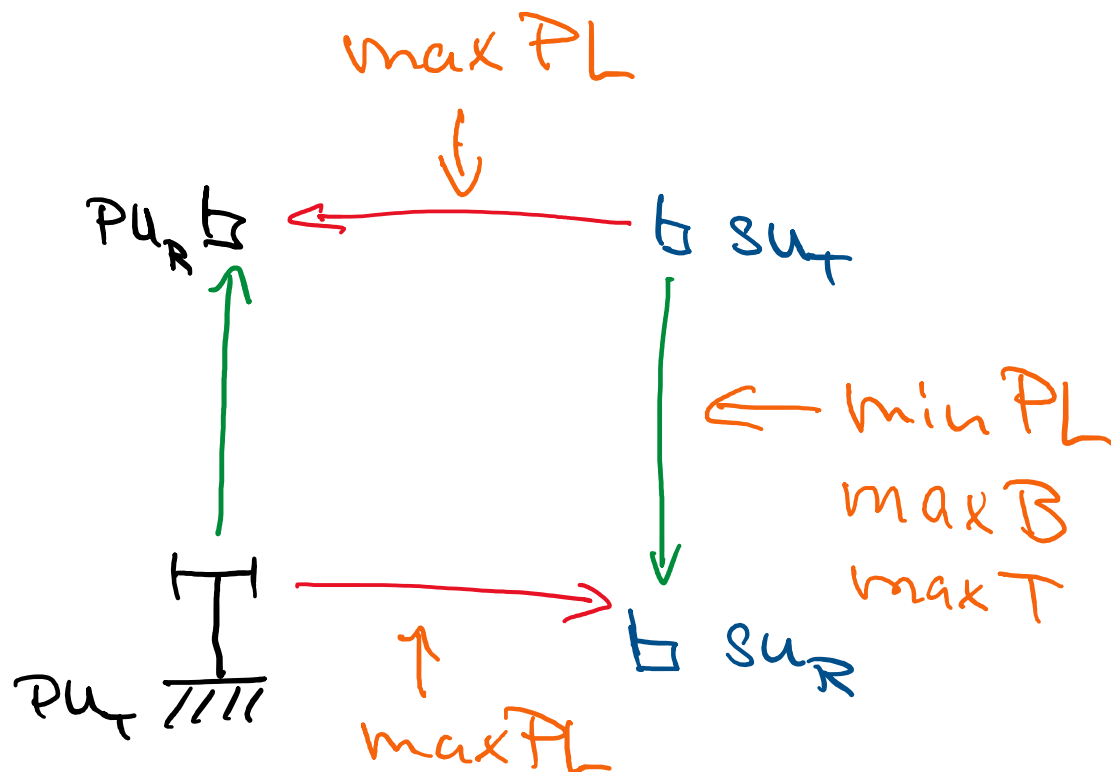
- Bandwidth of spectrum hole
- Path loss (interference generated) towards primary users
 - Some spectrum bands are more crowded than others
 - Based on the interference at primary receivers, the allowed sending power of secondary user can be derived → channel capacity is estimated
- Path loss between secondary users
 - Path loss is frequency dependent
 - Increased transmit power increases interference to primary network
- Characteristics of primary transmission
 - Radio technology (periodicity of signals, coding, detectability, etc.)
- Expected holding time of channel
 - Expected time duration the secondary user can occupy the channel before a spectrum handoff (handoff overhead)

Spectrum Decision – Optimization

Optimization problem:

Select spectrum hole where the primary user requirements are met along with the SU needs, including

- channel capacity (SINR and bandwidth B)
- expected holding time of channel T
- ...



Spectrum Mobility (Spectrum Handoff)

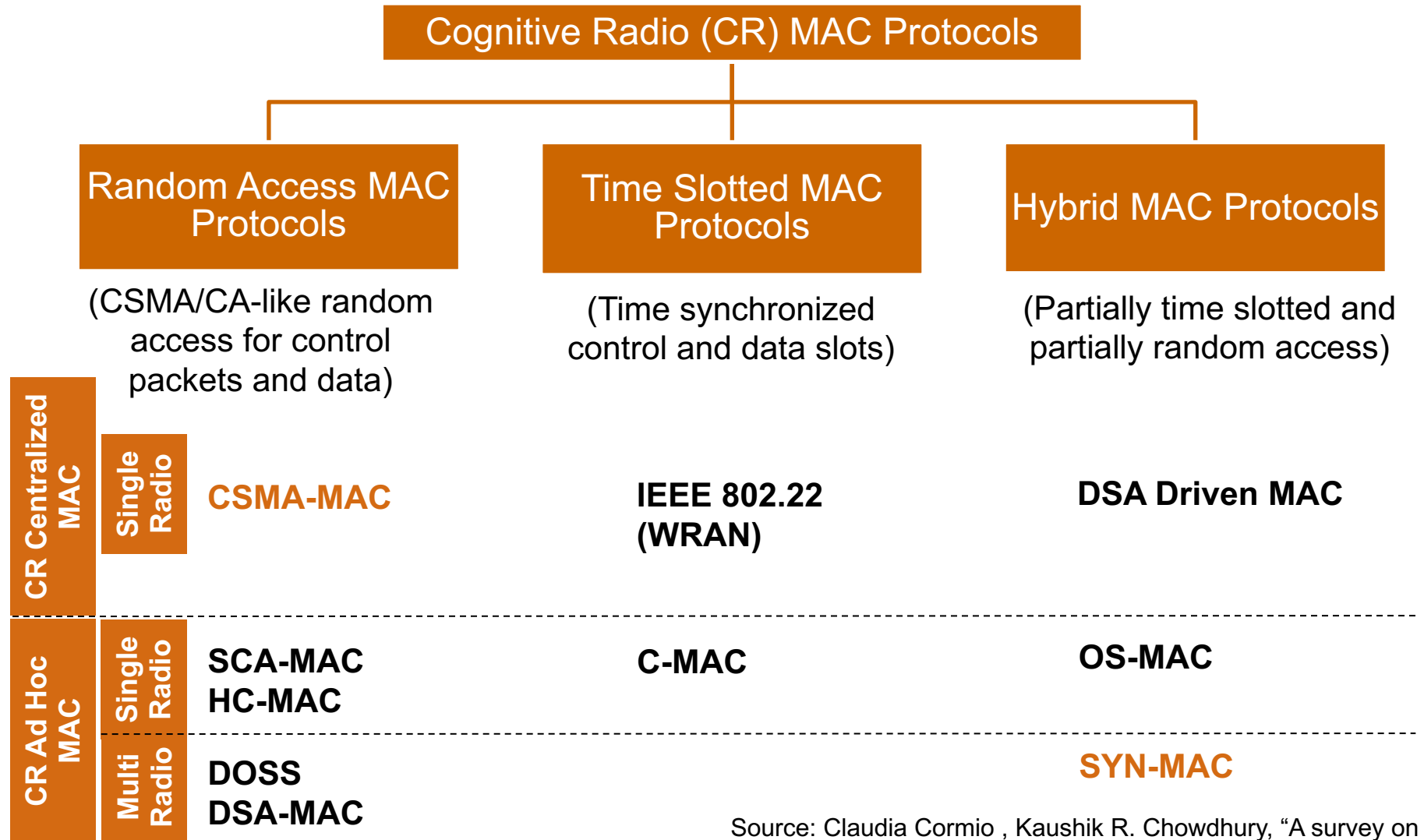
- The process when a secondary user changes its frequency of operation, also called spectrum handoff
- Reasons
 - Operating (secondary) channel becomes worse
 - Primary user returns to the channel
 - User movement (available spectrum bands change)
- Requirements
 - Low latency for vacation of spectrum
 - Transparency to upper layer protocols
 - No impairments of ongoing applications

Cognitive Radio MAC Protocols

Questions to be answered:

- What are the benefits of having multiple radios for cognitive stations?
- What are the advantages of centralized MAC schemes?
- What is the functionality of the common control channel? What are the challenges?
- What are the ideas of CSMA-MAC and SYN-MAC? How is the common control channel implemented in the two schemes?
- Identify the differences between CSMA-MAC and SYN-MAC and identify under which situations the individual schemes show their strengths

Cognitive Radio MAC Protocols – Classification



Source: Claudia Cormio , Kaushik R. Chowdhury, "A survey on MAC protocols for cognitive radio networks," Ad Hoc Networks journal, 2009

Cognitive Radio MAC Protocols – Classification

- According to the access method
 - Random access MAC protocols
 - No need for time synchronization
 - Based on Carrier Sense Multiple Access (CSMA)
 - Time slotted MAC protocols
 - Need for network-wide synchronization
 - Time is divided into slots for both control and data channels
 - Hybrid MAC protocols
 - Partially slotted transmission, in which
 - Signaling generally occurs over synchronized time slots
 - Data transmission may have random channel access schemes without time synchronization
- According to the architecture
 - CR centralized MAC
 - Central entity manages, synchronizes and coordinates operations among secondary users
 - CR Ad Hoc MAC
 - No central entity, neighbors cooperate to gain access to available channels
- According to the number of radios

Challenges of CR MAC Protocols

- Common control channel (CCC)
 - Tasks
 - Exchange of sensing information
 - Coordination (with a central entity) to organize the spectrum allocation
 - Transmitter-receiver handshake
 - Problems
 - Fixed CCC is infeasible (CCC must be vacated when a primary user appears on it)
 - CCC may vary over time and between users due to spatial distribution
 - If no CCC is allocated, transmitter-receiver handshake becomes a challenge
- Dynamic radio range
 - Radio range and characteristics change with operating frequency → CCC must be selected carefully (preferable to select CCC in lower spectrum bands)
- Spectrum unit
 - So far, we considered a channel as a basic spectrum unit (frequency space)
 - However, besides a frequency, a channel may be a recurring time slot, code, etc., as well, e.g. a time slot or code dedicated to the CCC

CSMA-based MAC (CSMA-MAC) Protocol

Idea:

- Operates in underlay (UL) or overlay (OL) spectrum sharing mode depending on PU activity
- Single transceiver required per user, CCC and data on same frequency
- Both PUs and SUs use CSMA to access the medium
- Extended sensing (waiting) period for SUs to prioritize PUs
- Power control for SUs to protect PUs from harmful interference (using CTS)
- SUs discriminate between 3 channel states:

Busy

- Busy state: PU is active, SU not allowed to transmit
 - » SU interference exceeds acceptable limit for PU
 - » Detected by failure of SU receiver to receive RTS

Partially
busy

- Partially busy state: PU is active, SU allowed to transmit in UL
 - » SU interference below acceptable limit for PU (UL mode)
 - » CTS reply to RTS from SU defines transmit power

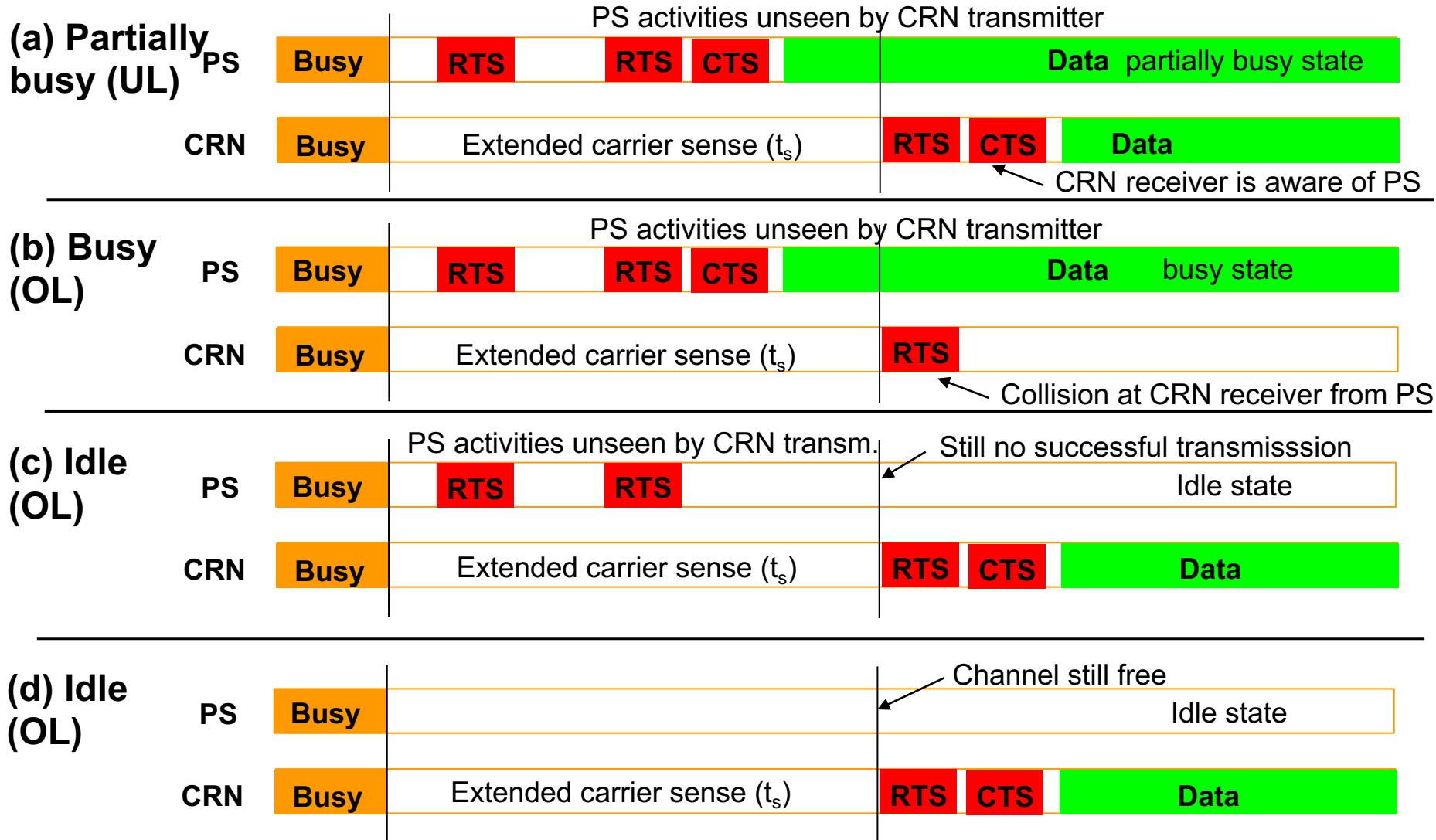
Idle

- Idle state: PU is idle, SU allowed to transmit in OL mode
 - » SU can transmit without power control (OL mode), no PU to be interfered with; SU maximizes its data rate

CSMA-MAC Protocol

Primary System (PS)

Cognitive Radio Network (CRN)

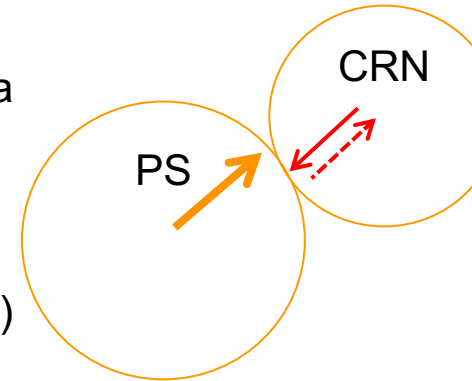


CSMA-MAC Protocol



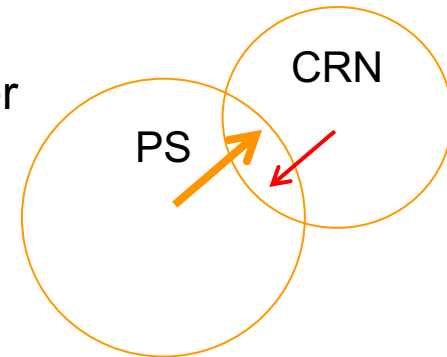
Case (a): CRN out of the range of PS (partially busy)

- Primary user gains the access to the channel and sends its data
- Secondary user senses the channel for an extended sensing period t_s
- The secondary user finds the channel vacant (assuming the primary and secondary users are separated by a large distance)
- The secondary user gains access to the channel and sends its data with reduced power as indicated by CTS



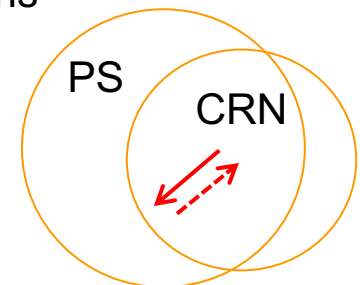
Case (b): CRN is in the range of PS but not of the PS transmitter (busy)

- RTS sent by the secondary user experiences collision
- The secondary user waits for the next transmission opportunity after repeating the previous sensing process



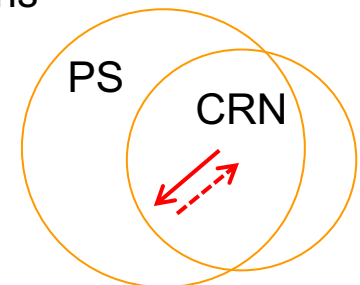
Case (c): PS experiences internal collision (idle)

- Primary user sends repeated RTSs all of them incurring collisions
- Secondary user can start transmission after t_s



Case (d): Channel is not used by PS (idle)

- Primary user has no packets to send, thus the channel is free
- Secondary user can start transmission after t_s

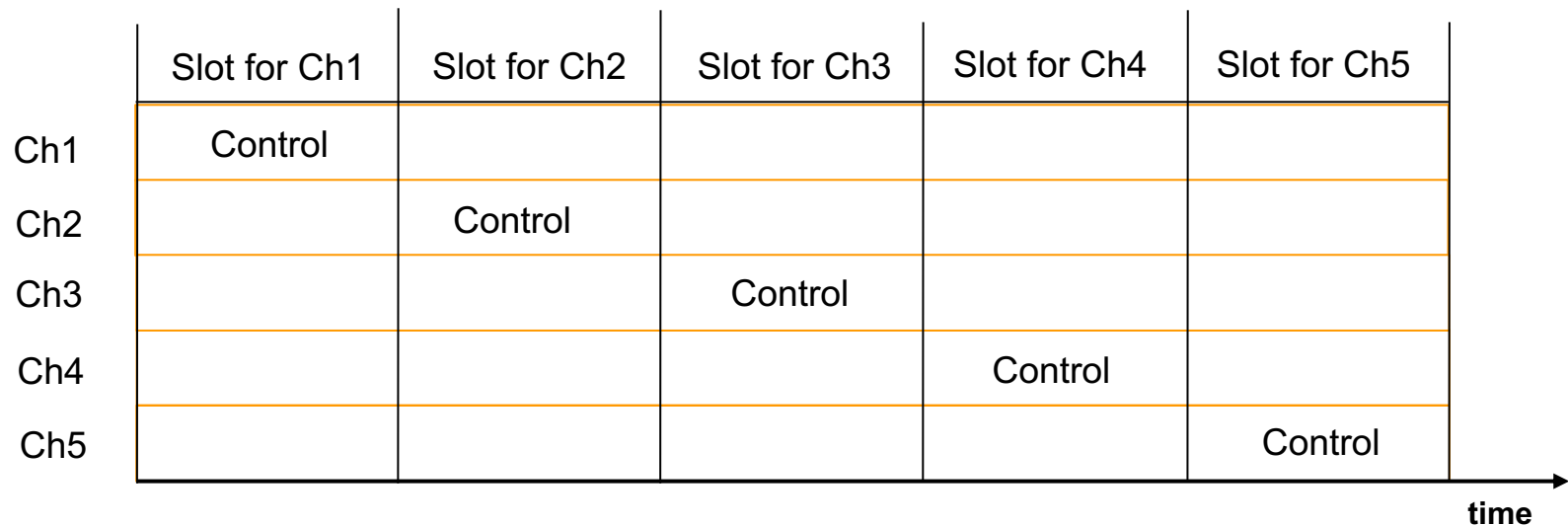


SYN-MAC Protocol

Idea:

- Each channel has a time slot dedicated to control messages (control slot)
- The control slot (CCC) is multiplexed in time between the different channels
- Each node has two radios, one for listening to control messages and one for transmission of data

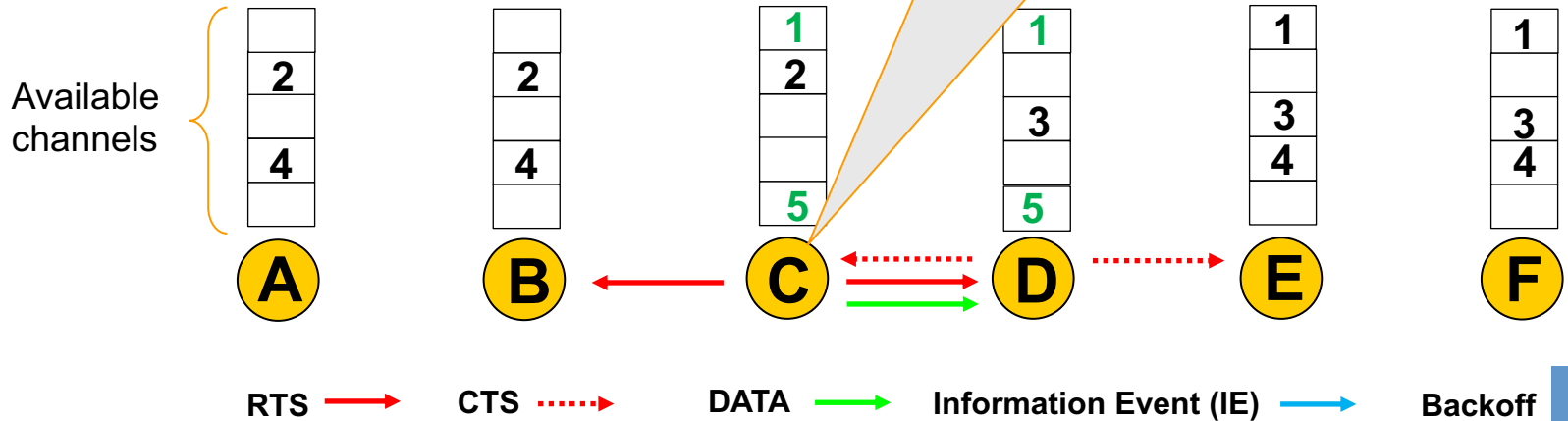
Time diagram



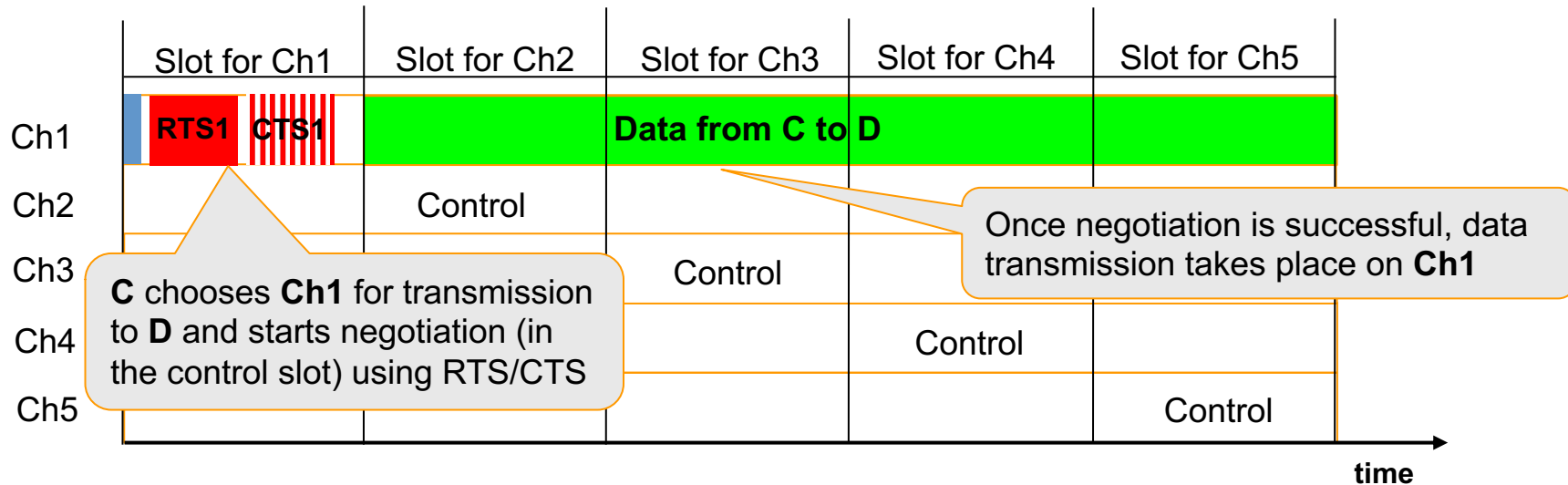
SYN-MAC Protocol – Example

C wants to send data to D
Channels 1 and 5 are common

Channel usage

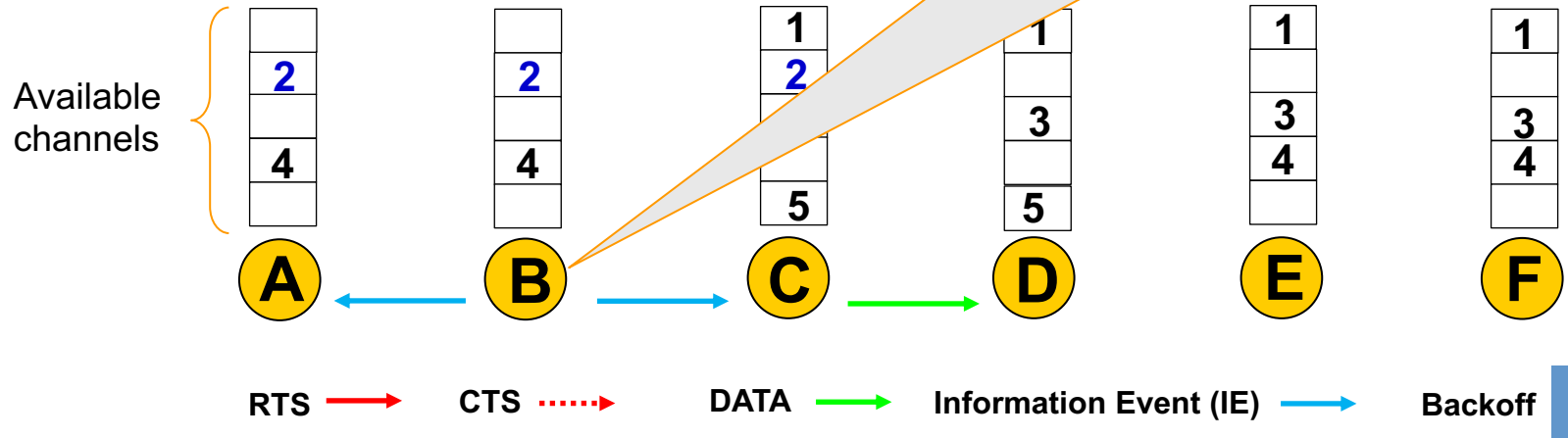


Time diagram



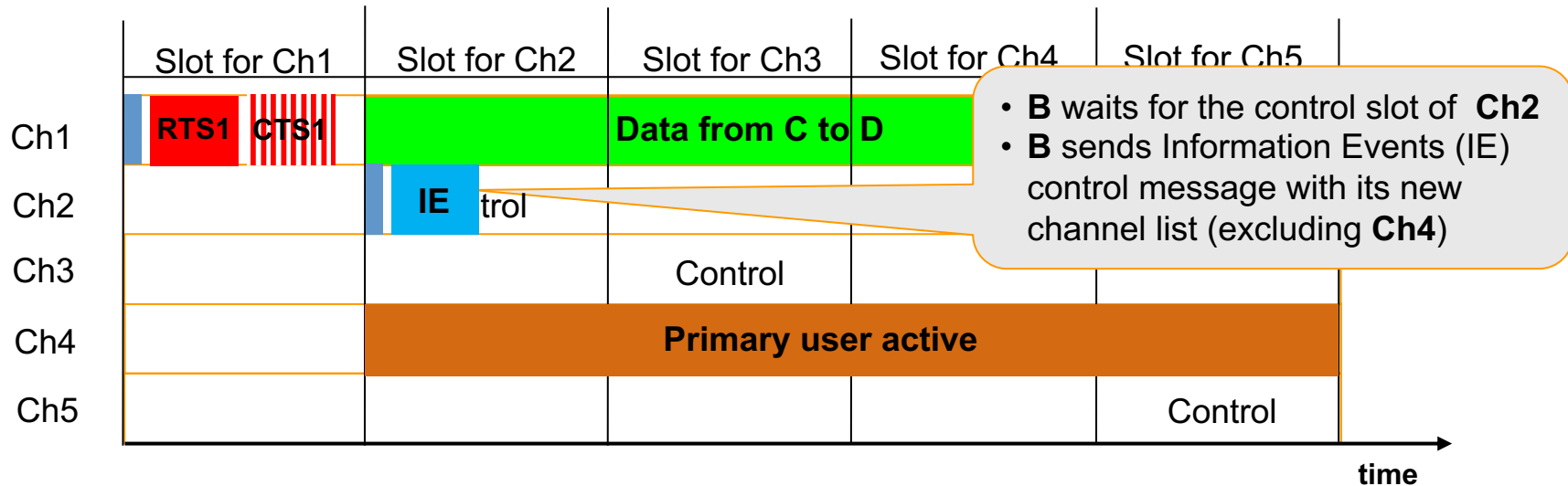
SYN-MAC Protocol – Example

Channel usage



- **B** observes that primary user of **Ch4** has returned
- **B** knows that it can reach its neighbors (**A** and **C**) through **Ch2**

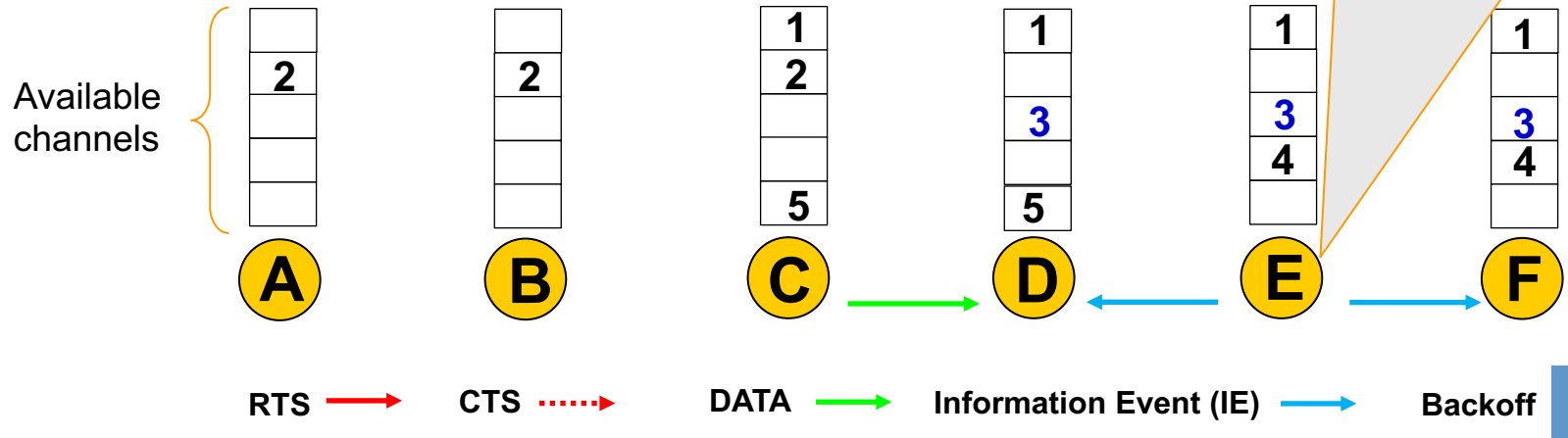
Time diagram



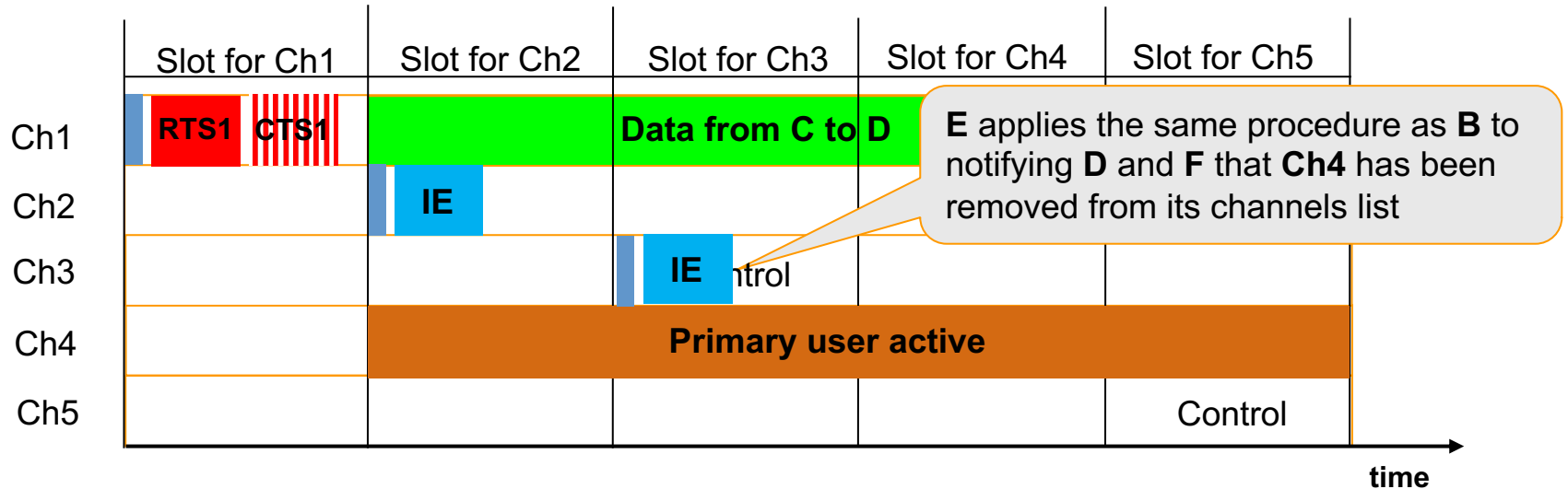
SYN-MAC Protocol – Example

- **E** observes that primary user of **Ch4** has returned
- **E** knows that it can reach its neighbors (**D** and **F**) through **Ch3**

Channel usage



Time diagram



CR Standards

IEEE 802.11af Super Wi-Fi (Feb. 2014)	IEEE 802.22 WRAN (July 2011)
TV white space spectrum in the VHF and UHF, 54 – 790 MHz, OFDM	
Cognitive sensing, flexible adjustment of bandwidth, modulation, power and coding	
Database driven approach: Access granted by Geolocation DataBase (GDB) Local sensing is used in certain special cases	
Up to 1 km, 35.6 Mbit/s for 8 MHz channels	30-100 km, 19 Mbit/s for 6 MHz channels
Access points (AP) and stations	Base stations (BS) and customer-premises equipment (CPE)
Wireless local area network (WLAN)	Wireless regional area network (WRAN)

Software Defined Radio (SDR)

Questions to be answered:

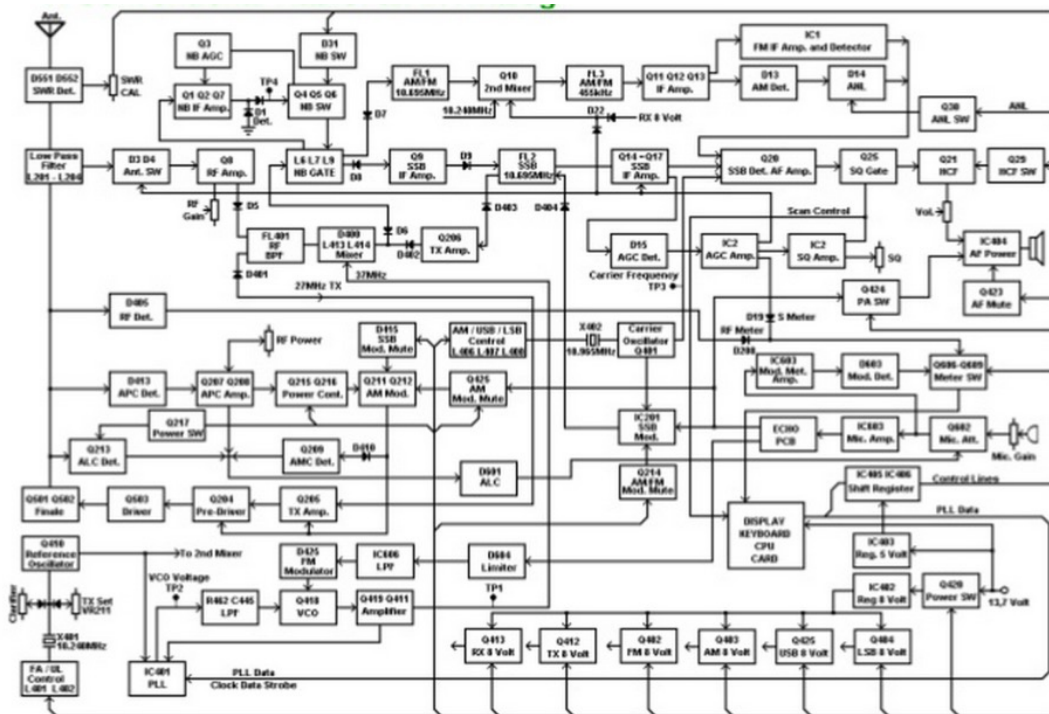
- What is the idea of SDRs?
- What are the advantages of SDRs
- What are the disadvantages and limits of SDRs?
- How do SDRs compare to traditional radio implementations?

Software Defined Radio – Practical Realization of a CR

- Requirements:
 - Capturing or sensing the radio environment
 - Reason about the observation
 - Adjusting own behavior accordingly
- Flexibility and reconfigurability

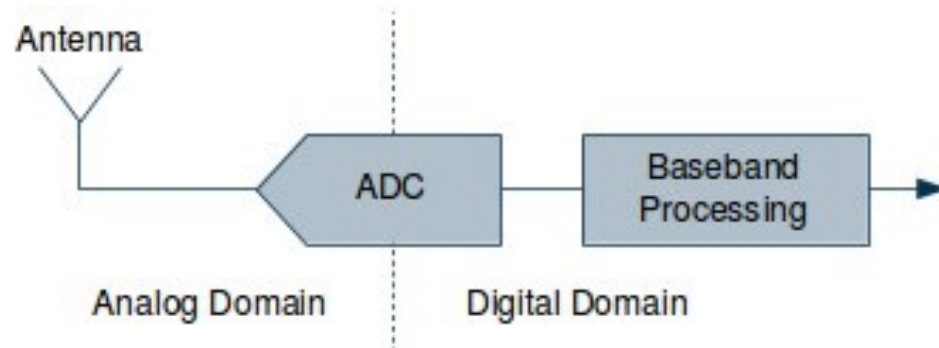
Traditional Radio Design

- Designed with concrete applications in mind
- Tailored to fixed requirements and operating conditions
- Highly optimized off-the-shelf and application-specific components
- Optimized for cost, energy-efficiency, and performance
- Static design: modifications impossible after fabricated

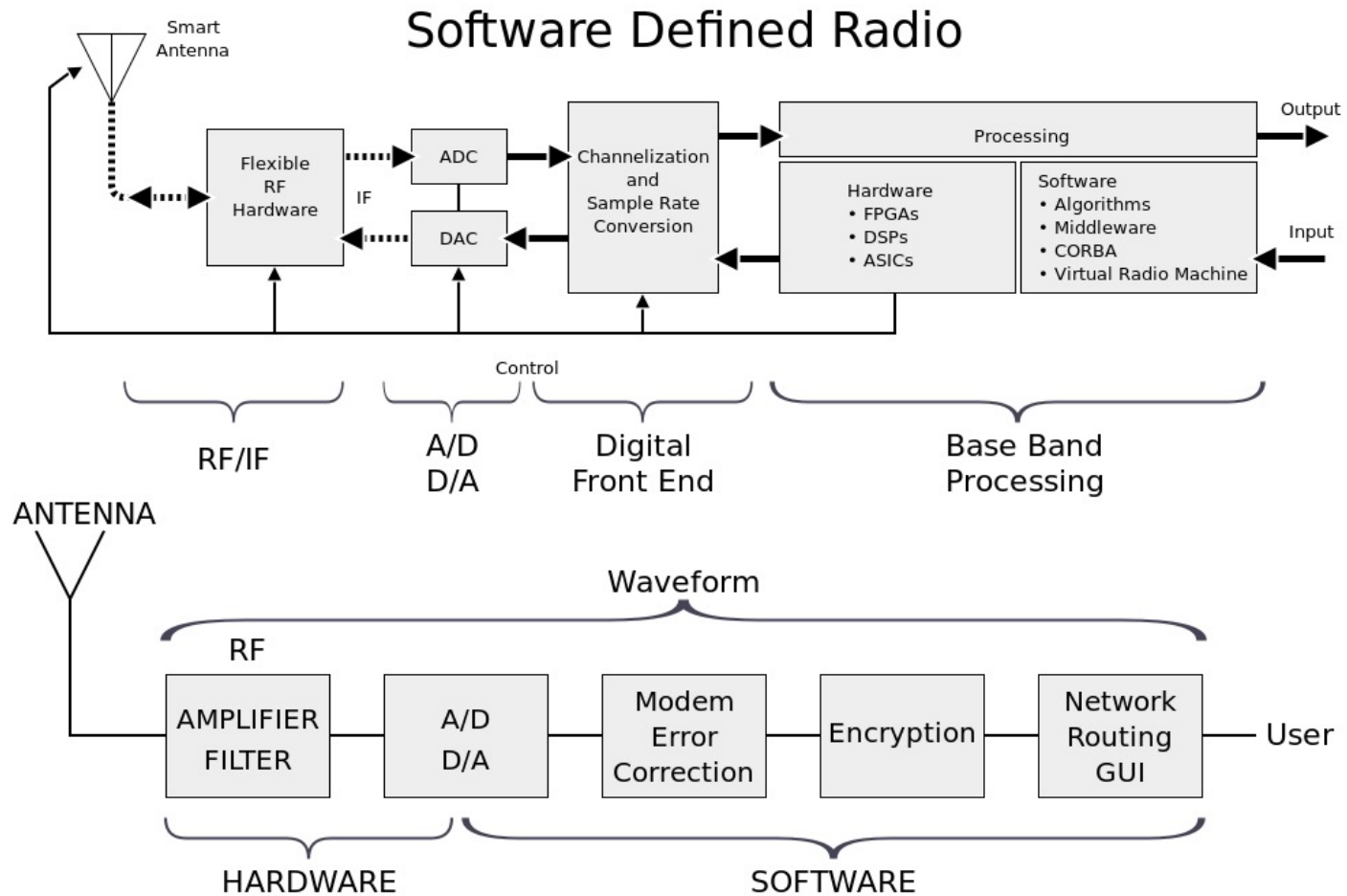


Idea: Software Defined Radio

- Reconfigurability by means of reprogrammable software
- Implement only most necessary things in HW
- Instead of HW components use DSPs, FPGAs, or GPPs
- Term Software Defined Radio also coined by Mitola
- Ideal SDR:



Practical Software Defined Radio



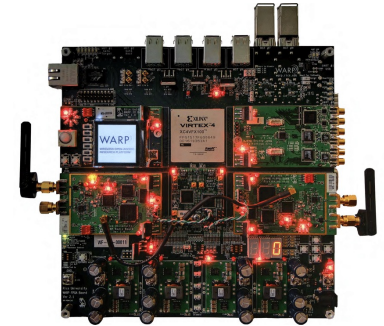
Source: Wikipedia

SDR Hardware

- Ranging between 10€ to 20000€
- Examples:
 - Ettus Research USRP family
 - HackRF
 - Blade-RF
 - Lime-SDR
 - RTL-SDR
 - SORA
 - WARP
 - DVB-T dongles
 - ...



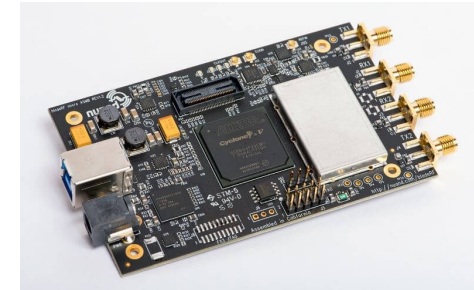
Lime-SDR



WARP



SORA



Blade-RF



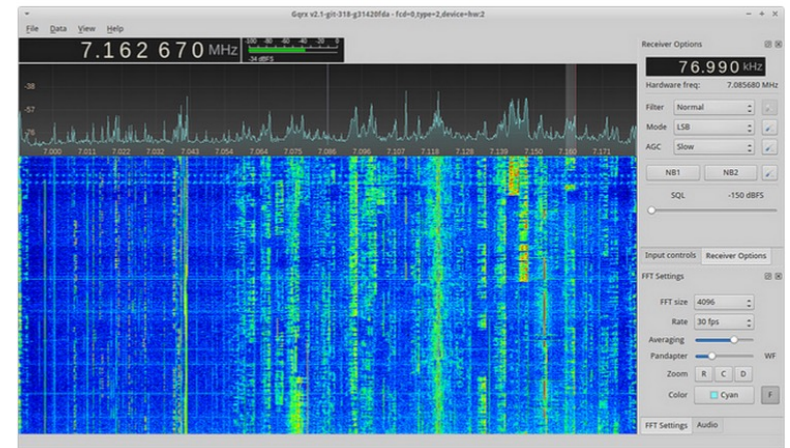
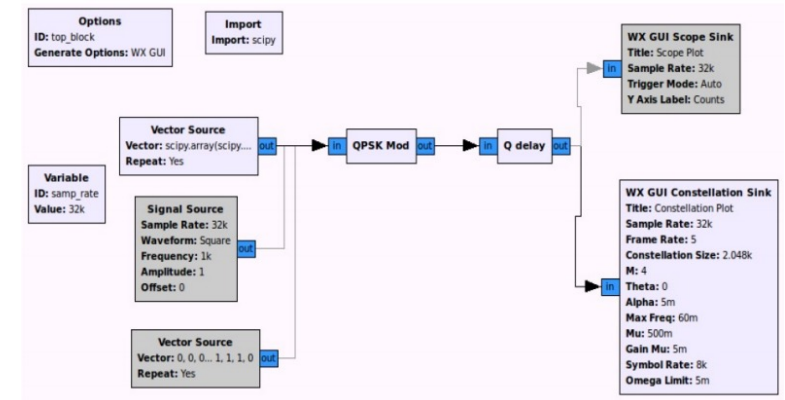
USRP



RTL-SDR

SDR Software and Applications

- GNU Radio (Python and C++)
- Iris (C++)
- Matlab
- LabVIEW by National Instruments
- Cellular Communications:
 - OpenBTS (GSM)
 - OpenAirInterface (LTE)
 - srsLTE
- Spectrum analyzer
 - gqrx



Conclusions

Cognitive radio technology enables the utilization of otherwise unused licensed spectrum

- Basic idea
 - Allocation of temporarily unused spectrum band
 - Utilization of the allocated band for communication as long as no primary user appears on this band and no (or limited) interference is produced for primary users in the neighborhood
- Requirements
 - Fast and accurate spectrum sensing equipment and mechanisms
 - Adequate spectrum decision approaches
 - Seamless spectrum mobility techniques
 - Efficient spectrum sharing methods
- IEEE 802.22, IEEE 802.11af are standards for cognitive radio networks

References

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